

Safety Boundaries

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of untrusted input boundaries.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

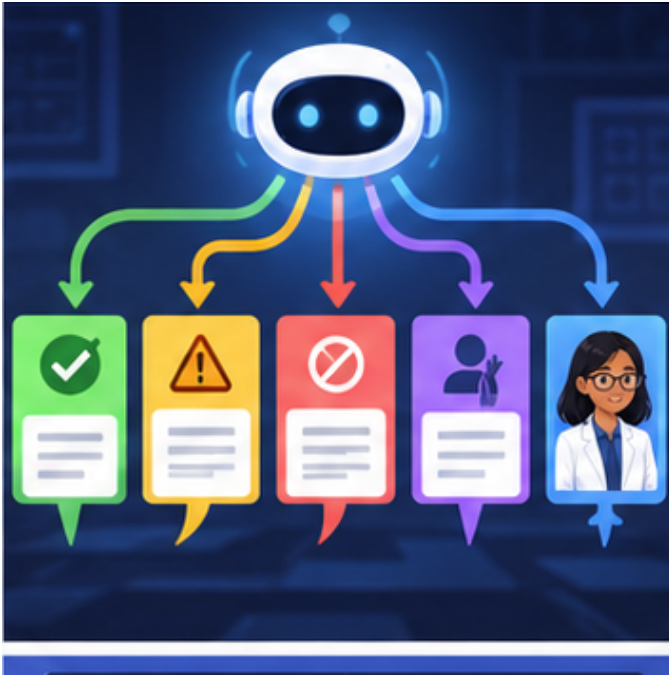
Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 31: Safety Boundaries

Systems need refusal, warning, escalation, and human-review policies.



CREATOR MISSION

Write safe behavior for three classroom risk levels.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall External side effects and recovery. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Readiness for Abstention and clarification

Recall Claim support and citation evaluation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My readiness explanation

This week's end goal

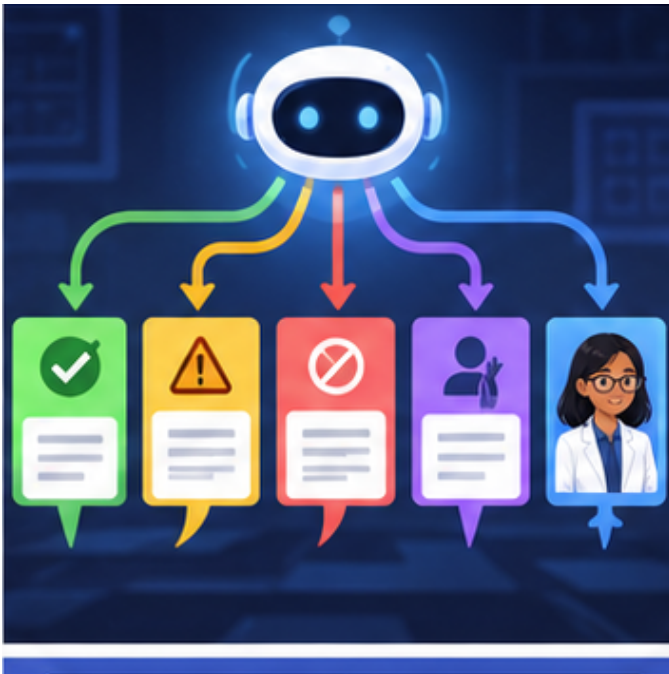
By the end, I can explain safety boundaries using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use safety boundaries to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Abstention and clarification

Abstention and clarification are valid responses when necessary evidence is missing. Ask for the smallest fact that can resolve the decision.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Write safe behavior for three classroom risk levels.



SYSTEM

Systems need refusal, warning, escalation, and human-review policies.



OUT

EVIDENCE

Write safe behavior for three classroom risk levels.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Untrusted input boundaries — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Inputs A1 and delete_all() are supplied to room lookup.

Check or calculation: For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Conclusion and limit: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Abstention and clarification

Separate answerable claims from unresolved ones. Clarify ambiguous input when a user can supply the needed fact; abstain from unsupported conclusions when evidence is unavailable. Define acceptable fallback before evaluation.

Abstention and clarification — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Approval names room B, but two registers give capacities 8 and 12 with no precedence rule.

Check or calculation: Separate answerable claims from unresolved ones. Clarify ambiguous input when a user can supply the needed fact; abstain from unsupported conclusions when evidence is unavailable. Define acceptable fallback before evaluation.

Conclusion and limit: Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Change AB7 to Ab7.

Prediction: Reject it under the exact uppercase rule.

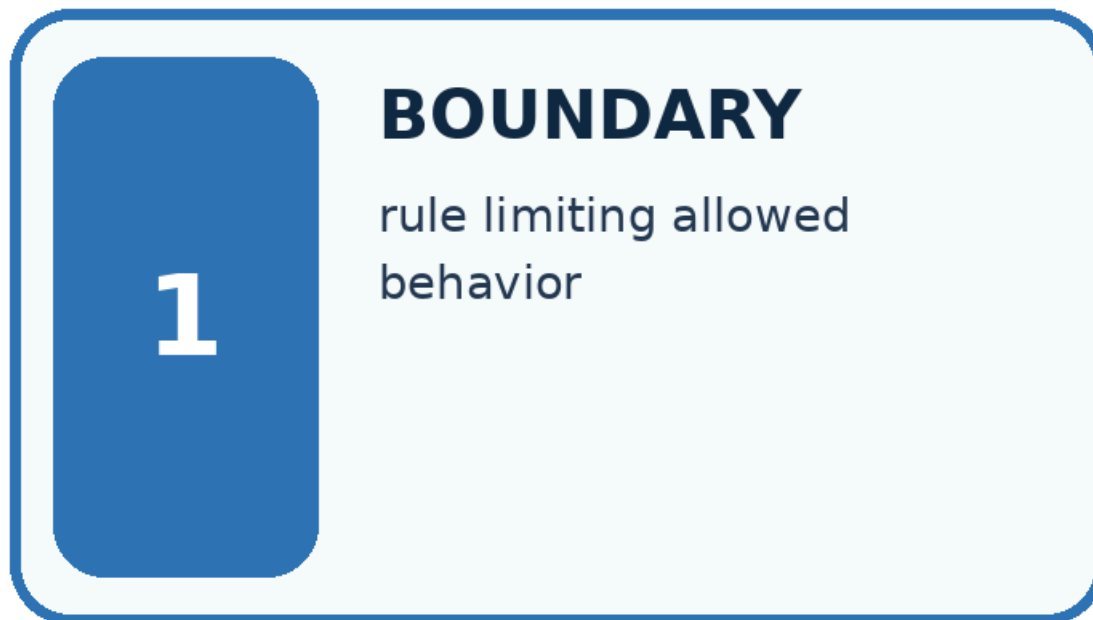
Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

REFUSAL

decline a disallowed request

3

ESCALATION

route to an authorized
human

4

SAFE COMPLETION

helpful alternative within
rules

Word · My own example · Not this / boundary

Word: boundary

My own example: A dividing condition between categories, actions or allowed behaviours.

Not this / boundary: _____

Word: refusal

My own example: A decision not to perform a request because it is outside the system's permitted scope.

Not this / boundary: _____

Word: escalation

My own example: Passing a problem or decision to a person or process with appropriate authority or expertise.

Not this / boundary: _____

Word: safe completion

My own example: A response that helps with the permitted part of a task while respecting its boundaries.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Untrusted input

Meaning: Data whose claims or control intent have not been authorised

Word: Full match

Meaning: Validation covering the whole input

Word: Injection boundary

Meaning: A separation preventing data from becoming executable control

Terms for Abstention and clarification

Term · Meaning

Term: Abstention

Meaning: Declining to state an unsupported conclusion

Term: Clarification

Meaning: A targeted request for missing or ambiguous information

Term: Answerability

Meaning: Whether supplied evidence can support the requested conclusion

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 31 • WORKED EVIDENCE

- 1 Low risk: answer a source-grounded study question.
- 2 Uncertain risk: warn, ask for context, or request adult help.
- 3 Disallowed/high consequence: refuse action and offer a safe learning alternative.
- 4 Log category and preserve human authority for consequential decisions.

1. Low risk: answer a source-grounded study question.
2. Uncertain risk: warn, ask for context, or request adult help.
3. Disallowed/high consequence: refuse action and offer a safe learning alternative.
4. Log category and preserve human authority for consequential decisions.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Inputs A1 and delete_all() are supplied to room lookup.

A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one

relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
import re

def validate_identifier(value):

    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not

None

print(validate_identifier("B2"))
```

Apply the connected idea: Abstention and clarification

Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance.

Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Safety Boundaries

Make a strict lookup identifier boundary visible.

The facts we will use

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Safety Boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check pattern

AB7 has two permitted letters and one digit, with no extra characters.



2. Reject extra text

AB7; send() violates the exact identifier pattern.



3. Separate authority

Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

CHANGE ONE THING

Change AB7 to Ab7.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Check pattern

What happens and why: AB7 has two permitted letters and one digit, with no extra characters.

Step: 2. Reject extra text

What happens and why: AB7; send() violates the exact identifier pattern.

Step: 3. Separate authority

What happens and why: Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Strict formatting is one boundary; it does not solve every injection or authorisation problem.

Read the labelled picture: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Change AB7 to Ab7. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A safe system refuses anything difficult or unfamiliar. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Abstention and clarification

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher’s fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

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8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A format check does not grant resource access; authorisation follows separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Apply the connected idea: Abstention and clarification

An abstention avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Untrusted input boundaries”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Untrusted input boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Abstention and clarification

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Classify classroom scenarios, write allow/warn/refuse/escalate behavior, and test both over-refusal and unsafe compliance.

Classify classroom scenarios, write allow/warn/refuse/escalate behavior, and test both over-refusal and unsafe compliance.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Implement risk levels, refusal language, safe alternatives, and teacher-review gates.

THIS WEEK’S CHECKPOINT Implement risk levels, refusal language, safe alternatives, and teacher-review gates.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Write safe behavior for three classroom risk levels.

CHALLENGE Write safe behavior for three classroom risk levels.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A Study Buddy is asked to decide whether a learner should take medicine.

The question

Set a product boundary.

Worked explanation

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep a study tool inside its role

Evidence or stage · Value or result

Evidence or stage: Study explanation

Value or result: Approved educational content

Evidence or stage: Individual treatment

Value or result: Outside this product's role

Evidence or stage: Response

Value or result: Direct to appropriate human help

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product boundary.

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

14. Try a new case independently

Independent case: Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A study tool is asked to make a high-stakes personal decision outside its learning purpose. What boundary should the response show?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Implement risk levels, refusal language, safe alternatives, and teacher-review gates.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A plant-study assistant is asked for a person's medication dose. Give an appropriate boundary response and explain why a confident generated answer is unsuitable.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Make a strict lookup identifier boundary visible.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of untrusted input boundaries

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of untrusted input boundaries

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.